Benefit: implicit bias

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Belief

Small initialisations (rich regime) yields an implicit bias towards *simple* estimators

Simple can refer to

- Small norm parameters $\|\theta\|_2$
- Sparse representation of the network (e.g. small number of kinks)

→ early alignment seems key to implicit bias:

only a few key directions are represented by the network after early alignment

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Drawback: reaching global minima is hard